

Abstract Diagrams for the Working Mathematician

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Abstract

Within the niche of mathematicians, there's a niche of those who need to draw diagrams in pure mathematics. Within *that* niche there's a niche of people who have experience in programming and feel comfortable writing code to produce graphics. This article is for them. It's a survey of the available tools and a showcase of a library I wrote specifically to simplify the process of drawing diagrams in differential geometry, topology, set theory, etc.

1 Introduction

Back in high school, I was attending a seminar on differential geometry, and I wanted to practice my $\text{\LaTeX}$ skills by typesetting the lecture notes. I found myself needing to draw simple surfaces such as those seen on Figure 1.

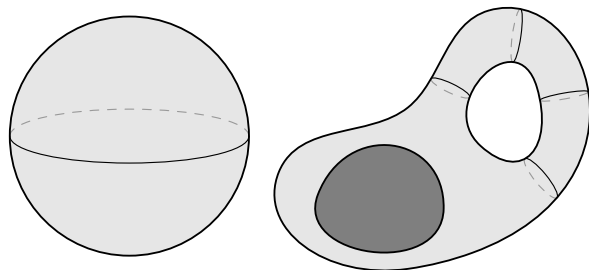


Figure 1: Basic two-dimensional manifolds

At that point I realized I wasn't sure how to draw this. I started by considering the available tools:

- **Inkscape** (<https://www.inkscape.org>)

This is a popular and versatile GUI program for drawing vectorized graphics, suitable when the diagram is simple enough to be drawn by hand with some precision assistance. This assumption holds for the left-hand side of Fig. 1, but not for the right-hand side. The reason is that each "cross section loop" has to be drawn at a precisely calculated position, where it meets its bounding paths at equal angles:

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